

Isaac Ivans

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GitHub: github.com/iivans | Portfolio: https://iivans.github.io

PROFESSIONAL SUMMARY

Computer Science graduate student and Graduate Assistant specializing in machine learning, computer vision, IT support, and systems networking. Experienced in developing deep learning models for medical imaging and deploying Raspberry Pi-based networking solutions. Strong background in troubleshooting, technical support, retail operations, staff training, and leadership.

TECHNICAL SKILLS

Languages: Python, C++, SQL, HTML

Machine Learning: TensorFlow, PyTorch, Deep Learning, Computer Vision, Image Processing

Data/Research: NumPy, Pandas, Scikit-learn, Matplotlib, MATLAB

Systems/IT: Linux, Windows, DNS, VPN, Raspberry Pi, POS Systems, Technical Support

Tools: Git, APIs, Microsoft Office, Zoom, Outlook

Professional: Leadership, Staff Training, Inventory Control, Payroll, Bookkeeping, Customer Support

PROJECTS

Deep-Learning Approaches for Automated Pneumonia Detection

Master's Thesis Project | TensorFlow, PyTorch, Python, Computer Vision

- Developed a comparative deep learning study for pneumonia detection using chest X-ray images.
- Designed and evaluated CNN, tuned CNN, VAE, and GAN-based classification approaches.
- Preprocessed medical images using grayscale conversion, resizing, normalization, and augmentation.
- Addressed class imbalance using class weighting and controlled evaluation methods.
- Evaluated models using accuracy, precision, recall, specificity, F1-score, ROC-AUC, and confusion matrices.
- Applied Grad-CAM visualizations to improve interpretability and examine model decision regions.
- Compared trade-offs between discriminative CNN models and generative-assisted approaches.

Pi-hole Network-Wide Ad Blocking + VPN Integration

Raspberry Pi Project | Linux, DNS, Networking, Tailscale/WireGuard

- Deployed Pi-hole on Raspberry Pi to provide network-wide DNS filtering and ad blocking.
- Configured DNS settings for local network devices and tested integration with Starlink internet.
- Troubleshoot DNS resolution issues, routing conflicts, service failures, and web interface errors.
- Configured secure remote access using VPN tools such as Tailscale/WireGuard.
- Explored uptime monitoring and reliability checks to detect outages and service problems.
- Strengthened hands-on experience with Linux administration, networking, and remote troubleshooting.

WORK EXPERIENCE

Graduate Assistant

New Mexico Highlands University | Las Vegas, NM | September 2024 - Present

- Assist professors with grading, instructional support, and course organization in computer science classes.
- Provide technical guidance to students working on programming and machine learning assignments.
- Support research initiatives involving artificial intelligence, software development, and computer vision.
- Provide remote technical support using Zoom, Microsoft Outlook, and university systems.
- Assist with workshops, training, documentation, and departmental records.
- Train new graduate assistants on departmental procedures and software tools.

Manager / Butcher

Grocery Outlet | Springer, NM | December 2018 - Present

- Manage daily grocery store operations including staff scheduling, inventory control, and vendor coordination.
- Train employees and support customer service, food safety, and operational efficiency.
- Oversee meat department operations, product preparation, and compliance procedures.
- Handle bookkeeping, payroll, documentation, and POS troubleshooting.
- Use IT skills to resolve system issues, support store technology, and maintain business operations.
- Coordinate with vendors and resolve customer inquiries in a fast-paced retail environment.

Work Study Student

New Mexico Highlands University | Las Vegas, NM | August 2020 - December 2020

- Provided technical support for computer labs and assisted faculty with administrative tasks.
- Troubleshot hardware and software issues for students, faculty, and university events.
- Helped maintain lab equipment and supported event setup/logistics.

Target Setter

NRA Whittington Center | Raton, NM | April 2019 - September 2019

- Prepared and maintained targets for shooting competitions while following safety procedures.
- Assisted event organizers with logistics, equipment setup, and customer service.

Cashier

Russell's Truck & Travel Center | Springer, NM | January 2016 - December 2018

- Processed customer transactions and provided front-line customer service.
- Maintained checkout area organization and assisted with stocking and inventory tasks.

EDUCATION

Master of Science in Computer Science

New Mexico Highlands University | Las Vegas, NM

May 2023 - Present

Bachelor of Science in Computer Science

New Mexico Highlands University | Las Vegas, NM

August 2018 - May 2023

CERTIFICATIONS AND LICENSES

Driver's License

ADDITIONAL INFORMATION

Authorized to work in the United States for any employer.

Language: English - Expert